

Anton Rehan Edirisinghe

he/him | rehane@umich.edu | +1 734-450-3623 | linkedin.com/in/rehan-ed | Ann Arbor, MI

Education

University of Michigan, College of Engineering -- Ann Arbor, MI Expected May 2027
B.S.E. Mechanical Engineering GPA: 3.68/4.00
Coursework: Mechanics of Materials, Dynamic Systems & Controls, Thermodynamics, DfM

Engineering Experience

Engineering Systems Inc. (ESi) – Engineering Consulting, Ann Arbor, MI May 2025 – Aug 2026
Engineering Consultant Intern – Marine & Automotive Design Group May 2026 – Aug 2026

- Built an AI agent that standardizes multi-page UD-10 crash reports into case summaries, cutting per-case processing from 30+ min to ~10 and flagging uncertain fields.
- Delivered a sensor-equipment tracking system (Co-Pilot, Power Automate, SharePoint, Teams) that replaced informal cross-office coordination and is now informing a company-wide rollout.
- Built a custom MATLAB to run regression analysis correlating accelerometer and XSens motion-capture data across 3 subjects and 12+ activities; designed the supporting database structure.

Engineering Consultant Intern May 2025 – Aug 2025

- Led an independent research project developing a methodology to track thrown-object motion from drone footage using photogrammetry, adding a new analysis capability.
- Supported active client cases through kinematic studies, Planar Impact Mechanics analysis, tire failure assessments, and experimental builds.
- Produced technical input for two expert reports used in legal proceedings, including a 30+ page report, plus on-site inspections and evidence documentation.

Perot Jain TechLab: Blueshift – U-M Center for Entrepreneurship Jan 2026 – May 2026
Student Consultant

- Advising a Boston cleantech startup in a 4-student team; built v1 of a Python and MATLAB process model of its leaching reactor, now preparing validation against experimental data.

Research & Sustainability

Global Carbon Initiative (GCI), University of Michigan Sep 2024 – Present
Research Assistant – Onboard Carbon Capture for Freight Rail (ME 490 RISE) Jan 2026 – Present

- Screened six point-source capture technologies against feasibility criteria and led a 25+ source literature review for a comparative LCA; targeting publication Nov 2026.
- Designed a proposed carbon-capture railcar architecture and built a first-pass life cycle inventory per tonne of CO₂ captured within railcar volume constraints.

Research Assistant – Solid-State Aluminum Recycling (ME 290 RISE) Sep 2024 – Dec 2024

- Investigated oxide fragmentation in roll-bonded AA1100, observing increased strength without alloying.

Leadership & Activities

Michigan Investment Group – Investment Analyst Feb 2024 – Dec 2024

- Ran LBO and DCF analyses and successfully pitched Blackstone Minerals (BSM) and Enphase Energy (ENPH), & Copper; selective admission (7.5% acceptance rate).

Michigan Supermileage – Mechanical Systems Engineer Sep 2025 – Present

- Designing aerodynamic surfaces and building Siemens NX molds for carbon fiber layouts.

Michigan Baja Racing – Machinist Sep 2024 – Aug 2025

- Machined custom vehicle components on the manual lathe and mill for the SAE Baja team.

Honors: International Climate Science Olympiad – Bronze medal, finalist (top 150 of 55,000).

Skills: MATLAB, AI Fluency Certified, Siemens NX, SolidWorks, DewesoftX DAQ, Power Automate, LCA

Interests: Michigan Squash Club (Sep 2024 – Present), wildlife and forensic photography

Work Authorization: 3 years of U.S. work eligibility under STEM OPT; no sponsorship required.